

I. feladat

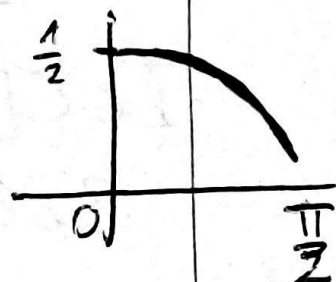
$$f(x) = \frac{\cos(x^2) + 1}{4}$$

$$x \in [0, \frac{\pi}{2}]$$

$$i.) \varphi: [0, \frac{\pi}{2}] \rightarrow [0, \frac{\pi}{2}] ?$$

$$\frac{\cos(x^2) + 1}{4} \leq \frac{1 + 1}{4} = \frac{2}{4} = \frac{1}{2}$$

$$\geq \frac{0 + 1}{4} = 0$$



$$[0, \frac{1}{2}] \subset [0, \frac{\pi}{2}]$$

$$ii.) \varphi'(x) = \frac{-\sin(x^2) \cdot 2x}{4}$$

$$|\varphi'(x)| = \left| \frac{\sin(x^2) \cdot x}{2} \right| \leq \frac{1 \cdot \frac{\pi}{2}}{2} = \frac{\pi}{4} < 1$$

$i.) + ii.) \xRightarrow{\text{BANACH}} \text{KONV.}$

Hibabeérték:

$$|x_k - x^*| \leq \left(\frac{\pi}{4}\right)^k \cdot \left(\frac{\pi}{2} - 0\right) = \frac{\pi}{2} \left(\frac{\pi}{4}\right)^k$$

II feladat

$$f(x) = \ln(x) + x^2 - 2 \quad x \in [1, 3]$$

$$a.) \quad f(1) = \ln(1) + 1^2 - 2 = -1 < 0$$

$$f(3) = \ln(3) + 9 - 2 > 8 > 0$$

$$f \in C[1, 3]$$



Bolzano $\implies \exists \xi \in [1, 3]$

b.)

$$X_{n+1} = X_n - \frac{\ln(x) + x^2 - 2}{1/x + 2x}$$

c.)

$$x \in [1, 3]$$

$$f'(x) = \frac{1}{x} + 2x > 0$$

$$f''(x) = 2 + \underbrace{\left(-\frac{1}{x^2}\right)}_{-1} \geq 1 > 0$$

$$f(x_0) \cdot f''(x_0) \geq 0 \iff f(x_0) > 0$$

$\iff x^* < x_0$ akkor Mon. növekvő

III feladat

$$f(x) = \sqrt{x+1} - x$$

-1	1	>	0	>	$\frac{-x'}{3 \cdot 4^2} = -\frac{1}{6}$	>	$\frac{-1 + 10 \cdot \frac{1}{6} \cdot \frac{1}{8^4}}{60}$
0	1	>	$-\frac{2}{3}$	>	$\frac{-\frac{6}{12} + 10 \cdot \frac{1}{8^5}}{15} \cdot \frac{1}{8^4} = -\frac{1}{60}$		
3	-1	>	$-\frac{4}{5}$	>			
8	-5	>		>			

$$N_3(x) = 1 + 0 \cdot (x+1) + \left(-\frac{1}{6}\right) \cdot (x+1) \cdot x + \frac{1}{60} x \cdot (x+1)(x-3)$$

$$|f(x) - N_3(x)| \leq \frac{M_4}{4!} \cdot |\omega(x)|$$

$$f'(x) = \frac{1}{\sqrt{x+1}} \cdot \frac{1}{2} - 1$$

$$f^{(iv)}(x) = \frac{-15}{16} \cdot \frac{1}{(x+1)^{7/2}}$$

$$f''(x) = \frac{-1}{\sqrt{(x+1)^3}} \cdot \frac{1}{4}$$

$$M_4 = \max_{x \in [0,8]} |f^{(iv)}(x)| = \frac{15}{16}$$

Csebenevher:

$$f'''(x) = \frac{+1}{(x+1)^{5/2}} \cdot \frac{3}{8}$$

$$M_3 = \frac{3}{8} \quad (x \in [0,8])$$

IV. feladat

$$\frac{2\sqrt{2}}{3} = \frac{\sqrt{2}}{3}$$

x_k	-1	0	1	2
y_k	1	-1	1	1

a) egyenes:

$$\begin{bmatrix} 4 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} p_0 \\ p_1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{array}{cc|c} 4 & 2 & 2 \\ 0 & 5 & 1 \end{array}$$

$$\begin{aligned} p_0 &= \frac{2}{5} \\ p_1 &= \frac{1}{5} \end{aligned}$$

$$P(x) = \frac{1}{5}x + \frac{2}{5}$$

b.) parabola:

$$\begin{array}{ccc|c} 4 & 2 & 6 & 2 \\ 2 & 6 & 8 & 2 \\ 6 & 8 & 18 & 6 \end{array} \rightarrow \begin{array}{ccc|c} 4 & 2 & 6 & 2 \\ 0 & 5 & 5 & 1 \\ 0 & 5 & 9 & 3 \end{array}$$

$$\begin{array}{ccc|c} 4 & 2 & 6 & 2 \\ 0 & 5 & 5 & 1 \\ 0 & 0 & 4 & 2 \end{array} \rightarrow \begin{aligned} p_0 &= -\frac{1}{10} \\ p_1 &= -\frac{3}{10} \\ p_2 &= \frac{1}{2} \end{aligned}$$

$$P(x) = \frac{1}{2}x^2 - \frac{3}{10}x - \frac{1}{10}$$

V. feladat

$$\int_0^1 \cos(\pi x^2) dx = ?$$

$$E(f) = (1-0) \cdot \cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$T(f) = \frac{1-0}{2} \cdot (\cos(0) + \cos(\pi)) = 0$$

$$S(f) = \frac{1-0}{6} \left(\cancel{\cos(0)} + 4 \cdot \cos\left(\frac{\pi}{4}\right) + \cancel{\cos(\pi)} \right) = \frac{\sqrt{2}}{3}$$

$$\left| \int_0^1 \cos(\pi x^2) dx - T_m(f) \right| \leq \frac{M_2}{12} \cdot (1-0)^3 \cdot \frac{1}{m^2}$$

$$M_2 = \max_{x \in [0,1]} |f''(x)| \leq 47$$

$$\sqrt{\frac{47}{12}} \cdot 10^2 < m$$

$$m = 20 \text{ már elég!}$$

$$f'(x) = -\sin(\pi x^2) \cdot 2x \cdot \pi$$

$$f''(x) = -2\pi \sin(\pi x^2) - 4\pi^2 x^2 \cos(\pi x^2)$$

$$|f''(x)| \leq 2\pi + 4\pi^2 \leq 47$$